

Business Engineering – Exercise Sheet: EOQ Models and Bullwhip Effect

Bachelor in Business Engineering

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Exercises on EOQ, EPQ, and Bullwhip Dynamics

Use exact expressions whenever convenient and round numerical answers to three decimals when needed. In Exercises 1–10, D denotes constant demand (units per year unless another time basis is stated), S the fixed order or setup cost, h the holding cost per unit per year, π the shortage cost per unit per year, Q the lot size, B the maximum backlog, $I_{\max}$ the maximum on-hand inventory, and p the production rate. In Exercises 11–20, $d(t)$ denotes the customer-demand deviation, $o(t)$ the order rate, $a(t)$ the arrival rate, $I(t)$ the inventory deviation, K the inventory-correction gain, and τ the replenishment lag. Use zero initial conditions throughout the bullwhip exercises.

When useful, use

$$\begin{aligned} TC_{\text{EOQ}}(Q) &= \frac{DS}{Q} + \frac{hQ}{2}, & TC_{\text{backlog}}(Q, B) &= \frac{DS}{Q} + \frac{h(Q-B)^2}{2Q} + \frac{\pi B^2}{2Q}, \\ TC_{\text{EPQ}}(Q) &= \frac{DS}{Q} + \frac{hQ}{2} \left(1 - \frac{D}{p}\right), & u(t) &= \text{unit step function}, \\ \dot{I}(t) &= a(t) - d(t), & \tau \dot{a}(t) + a(t) &= o(t), & o(t) &= d(t) - KI(t), \\ s^2 + 2\zeta\omega_n s + \omega_n^2, & \mathcal{L}^{-1} \left\{ \frac{1}{s+a} \right\} &= e^{-at}u(t), & \mathcal{L}^{-1} \left\{ \frac{1}{(s+a)^2} \right\} &= te^{-at}u(t), \\ & \mathcal{L}^{-1} \left\{ \frac{1}{(s+a)^2 + b^2} \right\} &= \frac{1}{b}e^{-at} \sin(bt)u(t). \end{aligned}$$

Part A – EOQ and EPQ models (Exercises 1–10)

1. **Ex 1:** A retailer of industrial labels faces constant demand

$$D = 24,000 \text{ units/year}, \quad S = 180 \text{ EUR/order}, \quad h = 6 \text{ EUR/unit/year}.$$

- (a) Write the total relevant annual cost function

$$TC(Q) = \frac{DS}{Q} + \frac{hQ}{2}$$

with the numerical parameter values inserted. (6 Points)

- (b) Compute the economic order quantity Q^* . (6 Points)

- (c) Determine the optimal number of orders per year and the optimal cycle length in working days if the firm operates 300 working days per year. (4 Points)
- (d) Determine the average inventory and the annual ordering and holding cost at Q^* . (4 Points)

2. **Ex 2:** A beverage bottler purchases pallet collars with constant demand

$$D = 12,000, \quad S = 90, \quad h = 3.$$

- (a) Determine the EOQ Q^* and the corresponding minimum relevant annual cost. (6 Points)
- (b) Compute the total relevant annual cost for

$$Q = 600 \quad \text{and} \quad Q = 1200.$$

(6 Points)

- (c) Which of the two candidate lot sizes is closer to the optimum in cost terms? Quantify the percentage deviation from the minimum cost in each case. (4 Points)
- (d) Determine the average inventory for $Q = 600$ and for $Q = 1200$. (4 Points)

3. **Ex 3:** Consider the classical EOQ model with constant demand, immediate replenishment, and no backlog:

$$TC(Q) = \frac{DS}{Q} + \frac{hQ}{2}.$$

- (a) Differentiate $TC(Q)$ with respect to Q and derive the first-order condition for the optimal lot size. (6 Points)
- (b) Show that the optimal lot size is

$$Q^* = \sqrt{\frac{2DS}{h}}.$$

(6 Points)

- (c) Compute the second derivative and show that the stationary point is a minimum. (4 Points)
- (d) Show that at Q^* , the annual ordering cost and the annual holding cost are equal. (4 Points)

4. **Ex 4:** An electronics wholesaler uses the classical EOQ model with

$$D = 36,000 \text{ units/year}, \quad S = 250 \text{ EUR/order}, \quad h = 5 \text{ EUR/unit/year}.$$

The firm operates 360 working days per year and the replenishment lead time is $L = 8$ days.

- (a) Compute the economic order quantity Q^* . (6 Points)
- (b) Determine the daily demand and the reorder point

$$R = dL.$$

(6 Points)

- (c) Suppose the holding cost falls to

$$h = 3.2 \text{ EUR/unit/year.}$$

Compute the new EOQ. (4 Points)

- (d) By what percentage does the EOQ change? Explain briefly why the change is smaller than the percentage change in h . (4 Points)

5. **Ex 5:** A service-parts depot allows planned backorders. The annual demand and cost parameters are

$$D = 18,000, \quad S = 120, \quad h = 4, \quad \pi = 16.$$

- (a) Write the total relevant annual cost function

$$TC(Q, B) = \frac{DS}{Q} + \frac{h(Q - B)^2}{2Q} + \frac{\pi B^2}{2Q}$$

with the numerical data inserted. (4 Points)

- (b) Compute the optimal lot size Q^* . (6 Points)
- (c) Compute the optimal maximum backlog B^* and the optimal maximum on-hand inventory

$$I^* = Q^* - B^*.$$

(6 Points)

- (d) Determine the cycle length and the fraction of the cycle with positive stock and with backlog. (4 Points)

6. **Ex 6:** Consider the EOQ model with immediate replenishment and planned backlog. Let I denote the maximum on-hand inventory and B the maximum backlog in one cycle, so that

$$Q = I + B.$$

- (a) Sketch one complete inventory cycle and explain why the average on-hand inventory over the full cycle equals

$$\frac{I^2}{2Q}.$$

(4 Points)

- (b) Show analogously that the average backlog over the full cycle equals

$$\frac{B^2}{2Q}.$$

(4 Points)

- (c) Starting from

$$TC(Q, B) = \frac{DS}{Q} + \frac{h(Q - B)^2}{2Q} + \frac{\pi B^2}{2Q},$$

derive the first-order condition with respect to B for fixed Q . (6 Points)

- (d) Show that the optimal split of the order quantity satisfies

$$B^* = \frac{h}{h + \pi}Q, \quad I^* = \frac{\pi}{h + \pi}Q.$$

(6 Points)

7. **Ex 7:** A distributor considers whether planned backorders are worthwhile. The data are

$$D = 10,000, \quad S = 200, \quad h = 5, \quad \pi = 20.$$

- (a) Compute the classical EOQ without backlog. (5 Points)
- (b) Compute the optimal lot size and the optimal maximum backlog if planned backorders are allowed. (7 Points)
- (c) Compute the minimum relevant annual cost in both cases. (4 Points)
- (d) Explain in one or two sentences why the backlog model typically recommends a larger lot size than the no-backlog EOQ model. (4 Points)

8. **Ex 8:** A factory produces filter cartridges internally. The data are

$$D = 30,000 \text{ units/year}, \quad S = 400 \text{ EUR/setup}, \\ h = 8 \text{ EUR/unit/year}, \quad p = 90,000 \text{ units/year}.$$

- (a) Write the EPQ cost function

$$TC(Q) = \frac{DS}{Q} + \frac{hQ}{2} \left(1 - \frac{D}{p}\right)$$

with the numerical values inserted. (4 Points)

- (b) Compute the optimal production lot size Q^* . (6 Points)
- (c) Determine the maximum inventory

$$I_{\max} = Q \left(1 - \frac{D}{p}\right)$$

and the average inventory at the optimum. (6 Points)

- (d) Determine the cycle length

$$T = \frac{Q}{D}$$

and the production uptime

$$t_p = \frac{Q}{p}.$$

(4 Points)

9. **Ex 9:** For the finite-production model with no backlog,

$$Q_{\text{EPQ}}^* = \sqrt{\frac{2DS}{h\left(1 - \frac{D}{p}\right)}}.$$

- (a) Show that as $p \rightarrow \infty$, the EPQ solution converges to the classical EOQ formula. (4 Points)

- (b) For

$$D = 16,000, \quad S = 150, \quad h = 2,$$

compute the classical EOQ and the EPQ for

$$p = 32,000 \quad \text{and} \quad p = 80,000.$$

(8 Points)

- (c) Determine the maximum inventory in both EPQ cases. (4 Points)

- (d) Explain how a faster production rate affects the recommended lot size and the average inventory. (4 Points)

10. **Ex 10:** An internal assembly line has annual demand

$$\begin{aligned} D &= 50,000 \text{ units/year}, & S &= 300 \text{ EUR/setup}, \\ h &= 4 \text{ EUR/unit/year}, & p &= 125,000 \text{ units/year}. \end{aligned}$$

Assume 250 working days per year.

- (a) Compute the optimal EPQ lot size. (6 Points)

- (b) Determine the production time per cycle and the non-production depletion time. (6 Points)

- (c) Determine the maximum inventory and the average inventory at the optimum. (4 Points)

- (d) If the production rate drops to

$$p = 60,000 \text{ units/year},$$

recompute the optimal lot size and state whether no-backlog operation is still feasible. (4 Points)

Part B – Bullwhip effect and replenishment dynamics (Exercises 11–20)

11. **Ex 11:** Consider the one-stage replenishment model

$$\dot{I}(t) = a(t) - d(t), \quad \tau \dot{a}(t) + a(t) = o(t), \quad o(t) = d(t) - KI(t).$$

(a) Take Laplace transforms and express $A(s)$ in terms of $O(s)$. (4 Points)

(b) Derive the transfer function

$$\frac{O(s)}{D(s)}.$$

(6 Points)

(c) Derive the transfer function

$$\frac{A(s)}{D(s)}.$$

(5 Points)

(d) Derive the transfer function

$$\frac{I(s)}{D(s)}.$$

(5 Points)

12. **Ex 12:** For the one-stage bullwhip model, the denominator of the demand-to-order transfer function is

$$\tau s^2 + s + K.$$

Compare it with the standard second-order form

$$s^2 + 2\zeta\omega_n s + \omega_n^2.$$

Consider the three parameter sets

$$\text{A: } K = 0.1, \tau = 1, \quad \text{B: } K = 0.25, \tau = 1, \quad \text{C: } K = 1, \tau = 2.$$

(a) Determine ω_n and ζ for all three cases. (8 Points)

(b) Classify each case as overdamped, critically damped, or underdamped. (4 Points)

(c) Rank the three cases by bullwhip oscillation risk. (4 Points)

(d) Which cases satisfy

$$K\tau > \frac{1}{4}?$$

Explain the significance of this inequality. (4 Points)

13. **Ex 13:** A replenishment stage uses

$$K = 0.25, \quad \tau = 1, \quad d(t) = 4u(t).$$

(a) Determine $D(s)$ and derive $O(s)$. (6 Points)

(b) By partial fractions, show that

$$O(s) = \frac{4}{s} + \frac{1}{\left(s + \frac{1}{2}\right)^2}.$$

(6 Points)

(c) Determine the time response $o(t)$. (4 Points)

(d) Determine the time and value of the peak order rate, and explain why this is bullwhip although the response is not oscillatory. (4 Points)

14. **Ex 14:** A replenishment stage uses

$$K = 0.5, \quad \tau = 2, \quad d(t) = 6u(t).$$

(a) Determine $O(s)$ and $I(s)$. (6 Points)

(b) Rewrite $O(s)$ and $I(s)$ into forms suitable for inverse Laplace transformation by completing the square. (6 Points)

(c) Determine the time responses $o(t)$ and $I(t)$. (4 Points)

(d) Explain why $o(t)$ shows bullwhip behavior and why $I(t)$ initially becomes negative after the demand step. (4 Points)

15. **Ex 15:** For the one-stage model,

$$\frac{O(s)}{D(s)} = \frac{(s + K)(\tau s + 1)}{\tau s^2 + s + K}.$$

(a) Set $s = j\omega$ and derive the magnitude formula

$$\left| \frac{O(j\omega)}{D(j\omega)} \right| = \frac{\sqrt{K^2 + \omega^2} \sqrt{1 + (\tau\omega)^2}}{\sqrt{(K - \tau\omega^2)^2 + \omega^2}}.$$

(8 Points)

(b) Evaluate the magnitude for

$$K = 0.8, \quad \tau = 3, \quad \omega = 0.4.$$

(4 Points)

(c) If the demand oscillation amplitude is 15 units/day, determine the order amplitude. (4 Points)

(d) State whether bullwhip occurs at this frequency. (4 Points)

16. **Ex 16:** Two replenishment policies are tested at the fluctuation frequency

$$\omega = 0.6 \text{ rad/day.}$$

Policy A uses

$$K = 0.3, \quad \tau = 1,$$

and Policy B uses

$$K = 1.0, \quad \tau = 3.$$

(a) Compute

$$\left| \frac{O(j\omega)}{D(j\omega)} \right|$$

for both policies. (8 Points)

(b) The customer-demand amplitude is 20 units/day. Determine the order amplitude under each policy. (4 Points)

(c) Which policy is more bullwhip-prone at this frequency? (4 Points)

(d) Relate the difference briefly to correction gain and lead-time lag. (4 Points)

17. **Ex 17:** At a certain fluctuation frequency, each of three similar supply-chain stages has amplification factor

$$|G(j\omega)| = 1.4.$$

(a) Determine the total amplification from customer demand to the third upstream order. (6 Points)

(b) If the customer-demand amplitude is 10 units/day, compute the order amplitude at the retailer, the wholesaler, and the manufacturer. (6 Points)

(c) Repeat the total-amplification calculation for four similar stages. (4 Points)

(d) Explain in one sentence why the expression

$$|G(j\omega)|^n$$

is important for multi-echelon supply-chain design. (4 Points)

18. **Ex 18:** At one fluctuation frequency, the stage-wise amplification factors from retailer to manufacturer are as follows.

Without point-of-sale demand sharing:

$$1.3, 1.4, 1.5, 1.6.$$

With point-of-sale demand sharing:

$$1.3, 1.1, 1.05, 1.02.$$

(a) Compute the total manufacturer amplification in both scenarios. (6 Points)

- (b) If the customer-demand amplitude is 12 units/day, compute the manufacturer order amplitude in both scenarios. (6 Points)
- (c) Compute the percentage reduction in manufacturer order amplitude due to demand sharing. (4 Points)
- (d) Explain briefly why better visibility reduces bullwhip. (4 Points)

19. **Ex 19:** A raw order signal is smoothed by the first-order filter

$$\frac{O_s(s)}{O_{\text{raw}}(s)} = \frac{1}{2s + 1}.$$

- (a) Determine the time constant and the DC gain of the smoothing filter. (4 Points)
- (b) Determine the amplitude ratio of the filter for a sinusoidal input at

$$\omega = 0.5.$$

(6 Points)

- (c) If the raw-order amplitude is 30 units/day, compute the smoothed amplitude. (4 Points)
- (d) For a raw-order step from 0 to 10, determine $o_s(t)$ and the time at which the smoothed signal first reaches 90% of its final value. (6 Points)

20. **Ex 20:** A firm currently uses the replenishment parameters

$$K = 1, \quad \tau = 4.$$

Two redesigns are considered:

$$\text{Option 1: } K = 1, \tau = 1.5, \quad \text{Option 2: } K = 0.4, \tau = 4.$$

Customer demand oscillates with amplitude 18 units/day at the frequency

$$\omega = 0.5.$$

- (a) Compute the damping ratio ζ for the current system and for both redesigns. (6 Points)
- (b) Compute

$$\left| \frac{O(j\omega)}{D(j\omega)} \right|$$

for the current system and for both redesigns. (6 Points)

- (c) Determine the order amplitude in all three cases. (4 Points)
- (d) Which redesign is more effective here? Justify your answer briefly using both the damping ratio and the amplification results. (4 Points)